CNC MACHINING CENTER – PREVENTIVE MAINTENANCE GUIDE

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1. Introduction

This guide defines the mandatory preventive maintenance schedule for all vertical and horizontal CNC machining centers. Proper adherence extends machine life, maintains accuracy, and reduces unplanned downtime.

2. Daily Maintenance Tasks

- Clean machine bed and remove all chips and swarf.
- Check coolant level and concentration (target 6–8% for most materials).
- Inspect tool holders and pull studs for damage or wear.
- Verify that all axis covers and way covers move freely.
- Check for unusual noises during spindle warm-up.

3. Weekly Maintenance Tasks

- Lubricate all linear guide ways and ball screws according to the lubrication chart.
- Clean or replace the coolant tank filter screen.
- Check hydraulic oil level (use only ISO VG 32 hydraulic oil).
- Inspect ATC (Automatic Tool Changer) gripper fingers and magazines.

4. Monthly Maintenance Tasks

- Measure spindle runout – maximum allowable is 0.005 mm at the taper.
- Check and adjust axis backlash if necessary.
- Clean the radiator of the spindle chiller and hydraulic unit.
- Verify the accuracy of the tool length measurement probe.

5. Critical Safety Rules

- Always place the machine in Emergency Stop mode before performing any tool change or maintenance inside the working area.
- Never use compressed air to clean the spindle taper – use only approved cleaning tools.
- Coolant must be replaced every 3–4 months or when pH falls below 8.5.
- Lockout-Tagout (LOTO) must be applied before opening any electrical cabinet.

6. Recommended Lubricants and Fluids

Component	Recommended Fluid	Specification
Linear Guides / Ball Screws	Way Oil	ISO VG 68
Hydraulic System	Hydraulic Oil	ISO VG 32
Spindle (if oil lubricated)	Spindle Oil	ISO VG 10 or 22
Coolant	Water-soluble coolant	6–8% concentration

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